

Build Your Own DIGITAL WEIGHING MACHINE

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This article describes how to build a simple and low-cost digital kitchen weighing machine, which can measure weight of up to five kilograms. This digital weighing machine is easier to design. For a heavy-duty weighing machine like your bathroom scale, just change the load cell or strain gauge. Everything else remains the same.

Circuit and working

Circuit diagram of the weighing machine is shown in Fig. 2. It is built around 5V voltage regulator 7805

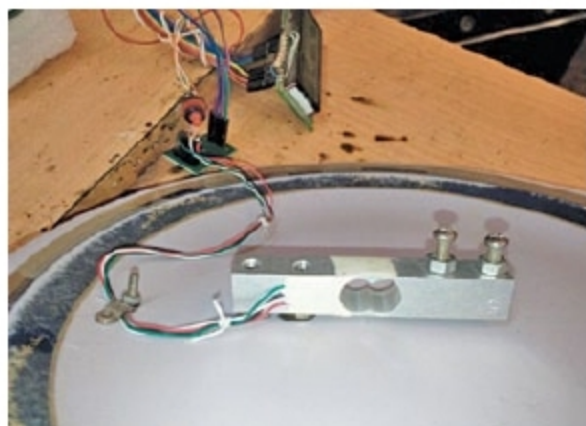


Fig. 1: Author's prototype

(IC1), analogue amplifier HX711 (IC2), ATmega328 microcontroller (IC3), load cell (strain gauge) for up to 5kg and a few discrete components. The author's prototype is shown in Fig. 1.

HX711 is used here because it gives a digital output, is low-cost and has an Adafruit header file for Arduino, which makes it easy to use. HX711 has input channel A (A+ and A-) and channel B (B+ and B-). Here, channel B is left unused. If you want to add one more load cell to take average, you can connect it to channels B- and B+. HX711, along with male bergstrip connectors, as shown in Fig. 3.

Load cell. Details of the load cell are shown in Fig. 4. The weighing scale has a metallic structure surrounded by strain-relieving frames, which pick up the strain when subjected to weight. Strain gauges are arranged in Wheatstone bridge configuration. Voltage is applied on one side, while the other side measures the change in voltage depending on the strain it suffers.

Load or weight is applied at the

free end. Strain gauges are applied on all three sides to pick up compressions and expansions. These are nothing but load signals that are processed and measured for weight measurements. Output is clubbed into four wires. Functions of the colour codes of the four wires are given in the table.

A yellow wire, if present, is the shield wire that must be connected to ground.

Software

The software consists of Arduino sketch (my_load_cell.ino), HX711 header files and other related libraries. First, add the library files (floatToString, HX711 and writeAnything) to Arduino IDE library and then run the sketch. To calibrate the probe, press the push button once. To reduce the number of readings for averaging, change the following script:

```
x = scale.get_units(25) × 45.3592 ;
//convert to gram for 25 readings
scale.read_average(5); //to read 5 average
```

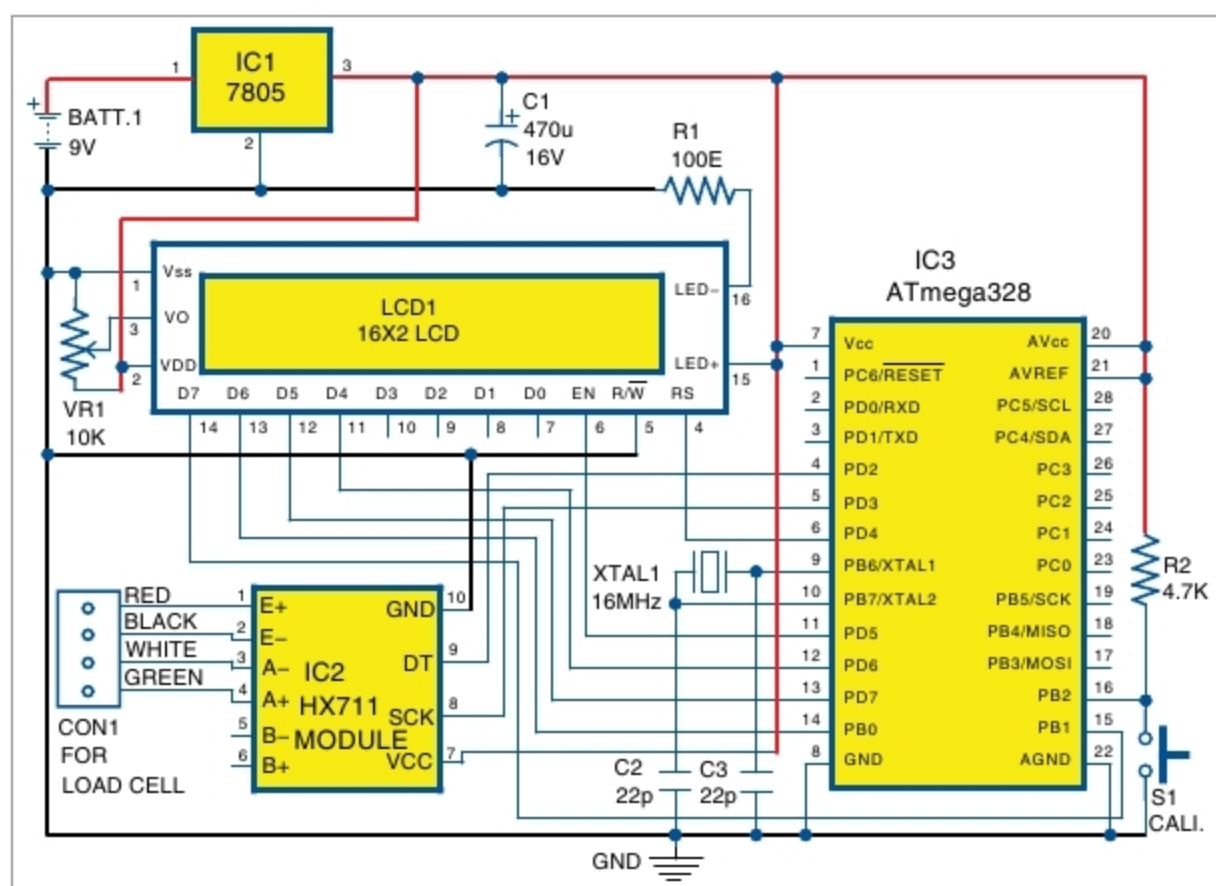


Fig. 2: Circuit diagram of weighing machine

FUNCTIONS OF COLOUR CODES OF WIRES

Colour code	Function	Connection to HX711
Red	+Excitation	E+
Black	-Excitation	E-
White	-Signal	A-
Green	+Signal	A+

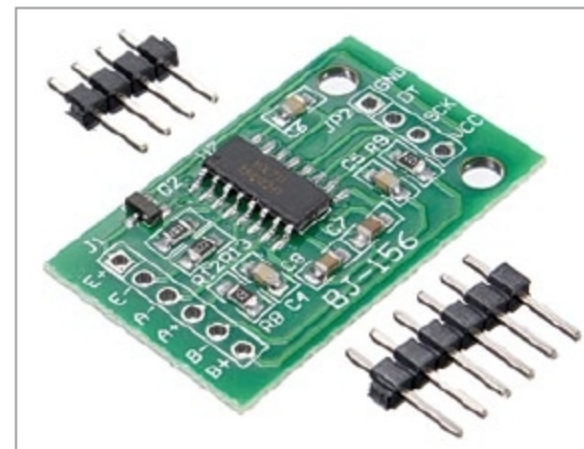


Fig. 3: Analogue amplifier HX711